



University of
Nottingham

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COMP4139

Machine Learning (MLE)

- Supervised Nonlinear Models (1)

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Learning Objective

- Understanding of the followings:
 - K Nearest Neighbour
 - Hard Margin SVM
 - Soft SVM
 - Kernel SVM
 - SVM Regression



Nonlinear Models

- Nonlinear algorithms assumes a nonlinear relationship between x and y . Thus, $f(x)$ can be a function of arbitrary complexity.
 - K nearest neighbour
 - Kernel SVM
 - Decision Trees
 - Neural Networks

K Nearest Neighbour (KNN)

Input: the training set T , category label set C and test object x . Each object has features x_1 and x_2 . The number of neighbours K .

Output: the category c_x of the test object x (c_x belongs to C)

begin

for each t belongs to T *do*

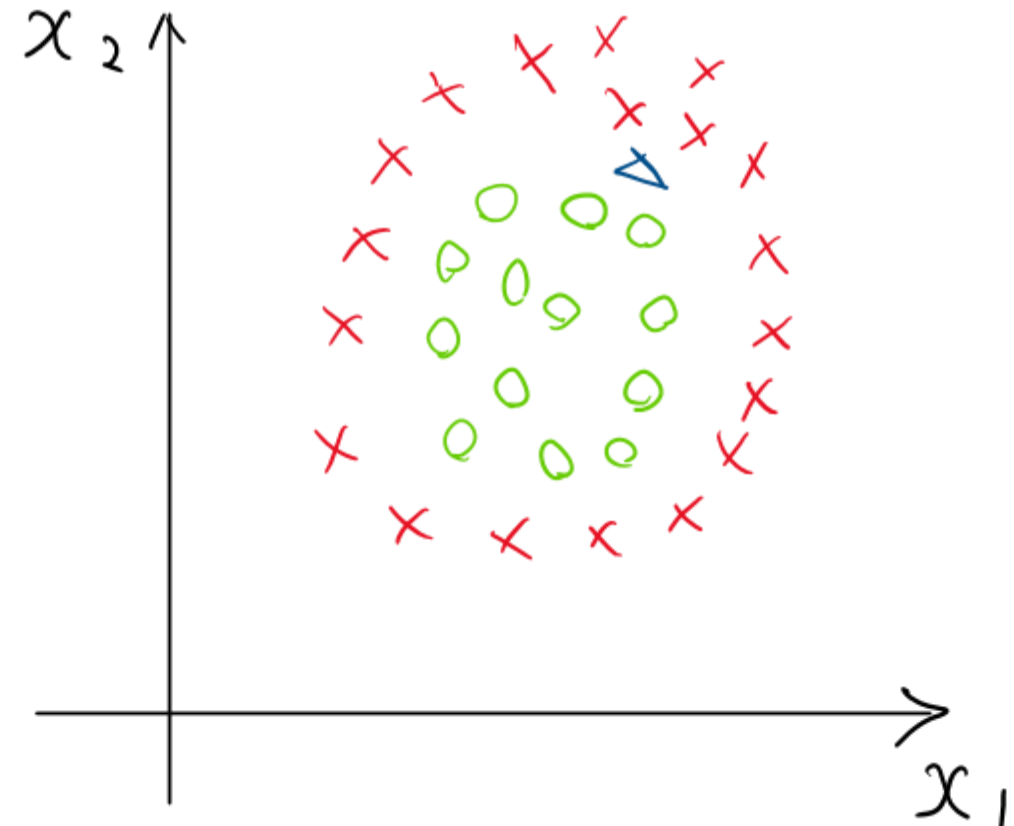
 calculate the distance $D(t, x)$

end for

 select K training examples from T that have the smallest distance D .

 calculate c_x as the majority class in the K examples.

end





Summary of KNN

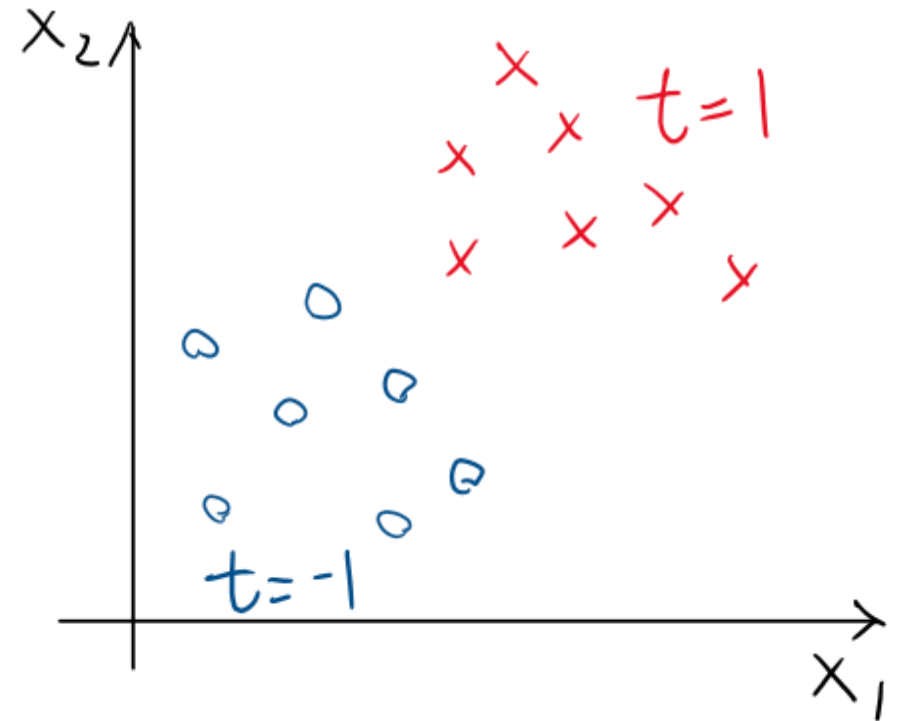
Step	Equation
Calculate distance $D(t,x)$	$D(t,x) = \sqrt{\sum_{i=1}^n (t_i - x_i)^2}$
Select K nearest neighbours	Identify K training samples $t_1, t_2, \dots, t_K$ with smallest $D(t,x)$
Predict majority class c_x	$c_x = \text{mode}(\{c_{t_k} \mid k = 1, 2, \dots, K\})$

- $D(t,x)$: Euclidean (or any other chosen) distance between training sample t and query sample x
- K : Number of neighbours to consider
- c_{t_k} : Class label of the k -th nearest neighbour

- What is the hyper parameter in KNN?
- How to determine K ?
- What happens if K is an even number and has an equal vote?
- Does KNN work for multi-class classification?
- What are the advantages and disadvantages of KNN?

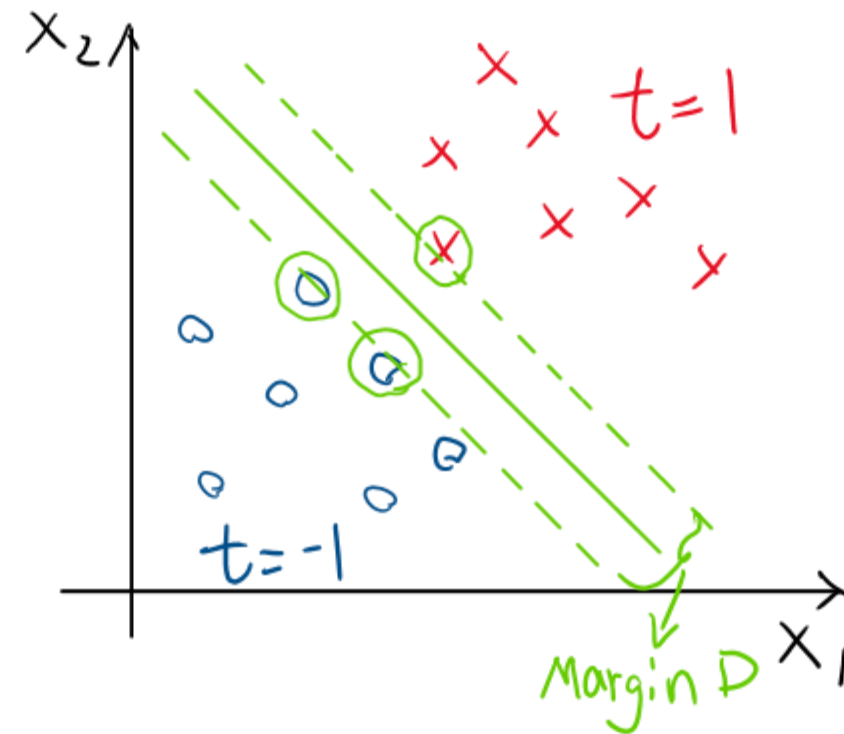
Hard Margin Support Vector Machines

- Hard margin SVM is a linear model: line in 2D, hyperplane in higher D.
- SVM seeks a decision boundary that maximises the **margin** between the two classes.



Hard Margin SVM

$n=(1, \dots, N)$ data points



Points on the decision boundary satisfy: $w^T x + b = 0$

Distance of a point x to the DB is: $\frac{w^T x + b}{\|w\|}$

Then we want all points x to satisfy:

$$\frac{w^T x + b}{\|w\|} \geq \frac{D}{2} \text{ for } t = 1, \quad \frac{w^T x + b}{\|w\|} \leq \frac{-D}{2} \text{ for } t = -1$$

For all points x_n , $n=(1, \dots, N)$:

$$\frac{t_n(w^T x_n + b)}{\|w\|} \geq \frac{D}{2}, \quad \text{then} \quad t_n(w^T x_n + b) \geq \frac{D\|w\|}{2}$$

By setting $\frac{D\|w\|}{2} = 1$, then $D = \frac{2}{\|w\|}$

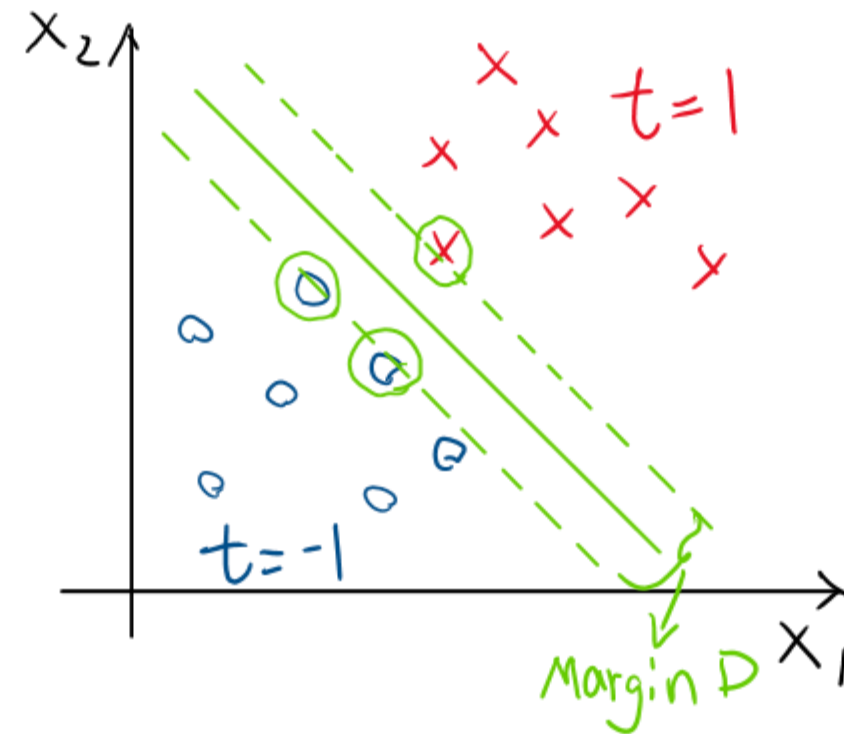
Then maximise D is equivalent to minimise $\|w\|^2$

Finally, $\min_{w,b} \frac{1}{2} \|w\|^2 \text{ s.t. } t_n(w^T x_n + b) \geq 1$



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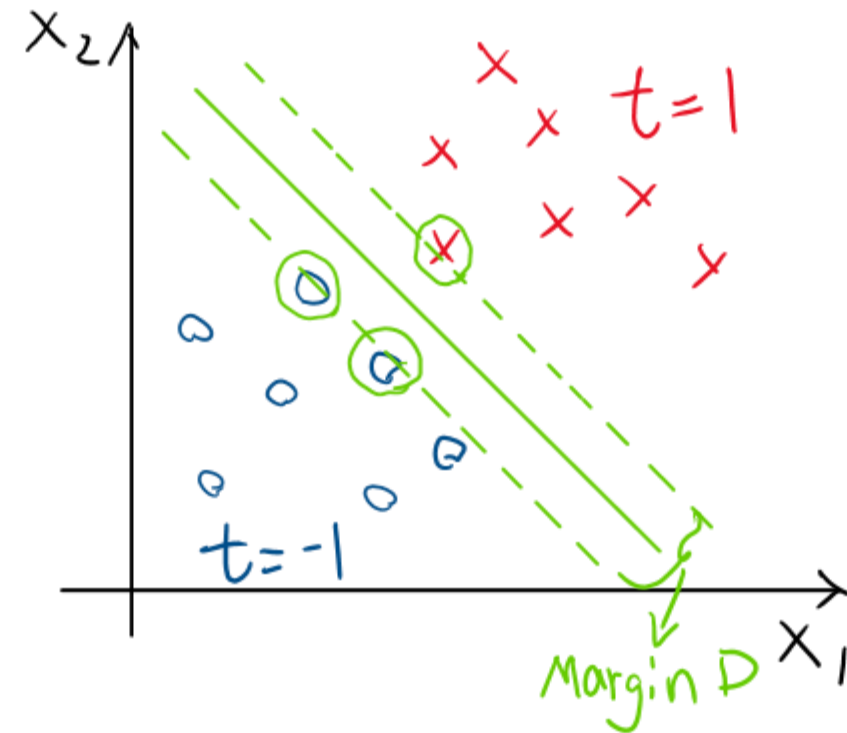
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Hard Margin SVM

$n=(1,\dots, N)$ data points



To find the DB, $\min_{w,b} \frac{1}{2} \|w\|^2$ s.t. $t_n(w^T x_n + b) \geq 1$

Optimised using Lagrange Multiplier:

$$L(w, b, a) = \frac{1}{2} w^T w - \sum_{n=1}^N a_n (t_n(w^T x_n + b) - 1)$$

The support vectors satisfy:

$$t_n(w^T x_n + b) = 1$$



Hard Margin SVM

Optimised using Lagrange Multiplier.

$$L(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_n (t_n (w^T X_n + b) - 1) \quad (1)$$

Set partial derivative = 0

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^N \alpha_n t_n X_n = 0 \Rightarrow w = \sum_{i=1}^N \alpha_n t_n X_n \quad (2)$$

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^N \alpha_n t_n = 0 \quad (3)$$

Substitute (2) (3) into (1), and optimise α .

$$\min_{\alpha} \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N t_n t_m \alpha_n \alpha_m X_n^T X_m - \sum_{n=1}^N \alpha_n$$

y	$\partial y / \partial w$
Aw	A^T
$w^T A$	A
$w^T A w$	$A w + A^T w$



Hard Margin SVM

$$\min_{\alpha} \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N t_n t_m \alpha_n \alpha_m X_n^T X_m - \sum_{n=1}^N \alpha_n$$

Use quadratic solver:

$$\min_{\alpha} \frac{1}{2} \alpha^T \underbrace{\begin{bmatrix} t_1 t_1 X_1^T X_1 & t_1 t_2 X_1^T X_2 & \dots & t_1 t_N X_1^T X_N \\ t_2 t_1 X_2^T X_1 & t_2 t_2 X_2^T X_2 & \dots & t_2 t_N X_2^T X_N \\ \vdots & \vdots & \dots & \vdots \\ t_N t_1 X_N^T X_1 & t_N t_2 X_N^T X_2 & \dots & t_N t_N X_N^T X_N \end{bmatrix}}_{\Theta} \alpha + (-1^T) \alpha$$

pair-wise dot product

Training data $N \times N$

$$\Theta = \frac{1}{2} \hat{\alpha}^T \Theta \hat{\alpha} + (-1^T) \hat{\alpha}$$

solve α , we can get $w = \sum_{i=1}^N \alpha_i t_i X_i$, use $t_n (w^T X_n + b) = 1$ to solve b
(support vectors $\alpha > 0$)

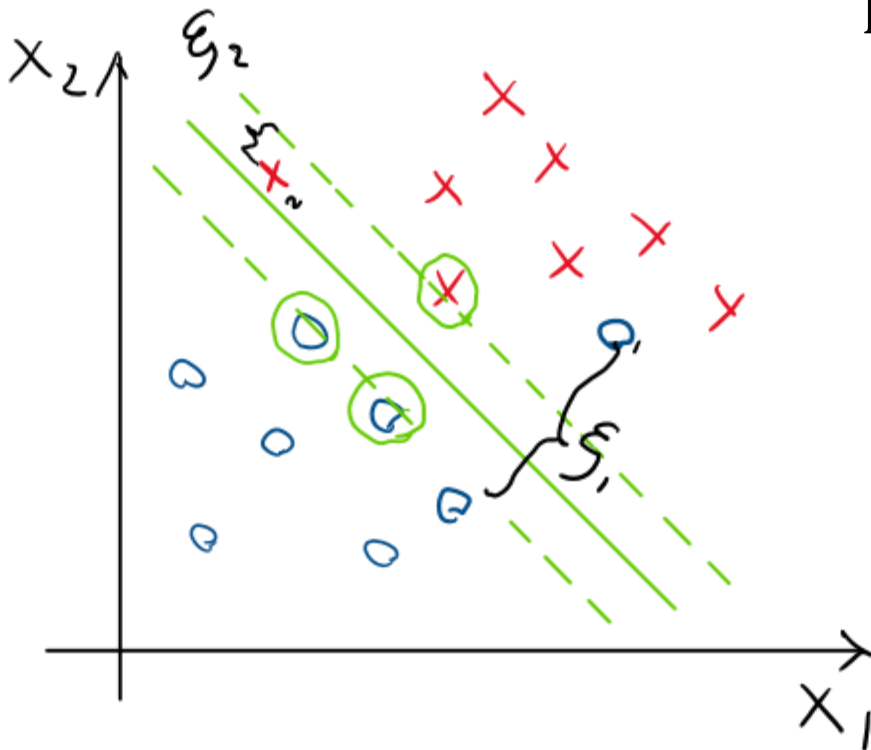


Summary of the Hard Margin SVM

Concept	Formula	Meaning
Primal objective	$\left(\min_{w,b} \frac{1}{2} w ^2 \right) s.t. (t_n(w^T x_n + b) \geq 1)$	Maximize margin
Lagrangian	$\left(L(w, b, a) = \frac{1}{2} w^T w - \sum_n a_n [t_n(w^T x_n + b) - 1] \right)$	Combine objective + constraints
Stationary conditions	$\left(w = \sum_n a_n t_n x_n, \sum_n a_n t_n = 0 \right)$	From partial derivatives
Dual problem	$\left(\min_a \frac{1}{2} \sum_n \sum_m a_n a_m t_n t_m x_n^T x_m - \sum_n a_n \right)$	Quadratic form
Matrix version	$\left(\min_a \frac{1}{2} a^T Q a - 1^T a \right) with (Q_{nm} = t_n t_m x_n^T x_m)$	What's fed into a QP solver
Solution	$\left(w = \sum a_n t_n x_n, b = t_n - w^T x_n \right) \text{ for support vectors}$	Construct classifier

Soft Support Vector Machines (Linear Model)

- Soft SVM allows some points to be on the wrong side of DB.



Hard margin SVM: $\left(\min_{w,b} \frac{1}{2} \|w\|^2 \right) \text{ s.t. } (t_n(w^T x_n + b) \geq 1)$

Allow some violations,

ξ_n : slack that measures the violation for x_n

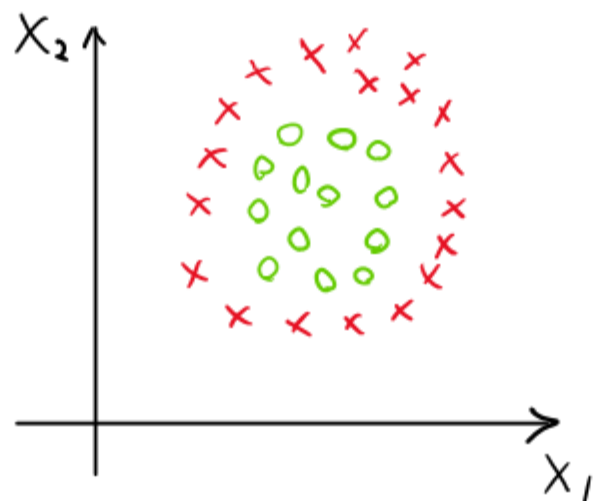
Then we have the soft SVM,

$$\min_{w,b,\xi_n} \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n \text{ s.t. } t_n(w^T x_n + b) + \xi_n \geq 1$$

C is a weighting term: *higher C results in smaller D*

Kernel Support Vector Machines (Nonlinear)

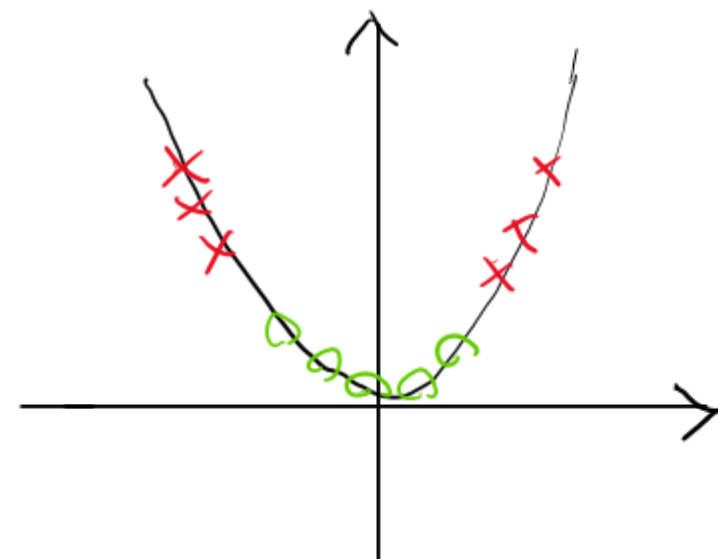
Not linear separable



Map it to a higher
dimensional space



Becomes linearly separable
 $\phi(x) = (x, x^2)$





Kernel SVM

- SVM optimisation requires dot product of all pair-wise samples (e.g. 5 samples, requires 10 pairs of dot product)
- Simply increase the dimensionality will increase the computation significantly.
- Hence, we use a kernel trick!

Two original feature x_1, x_2

Use quadratic function map to higher dimension:

$$\phi = (x_1, x_2, x_1^2, x_2^2, x_1x_2)$$

Play a trick to convert it to:

$$\phi = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$$

For a pair of sample x_i, x_j , the dot product is:

$$\phi(x_i) \cdot \phi(x_j) = (1 + x_1^i x_1^j + x_2^i x_2^j)^2 = (1 + x^i \cdot x^j)^2$$

Dot product of the original feature



Polynomial Kernel and Gaussian Kernel

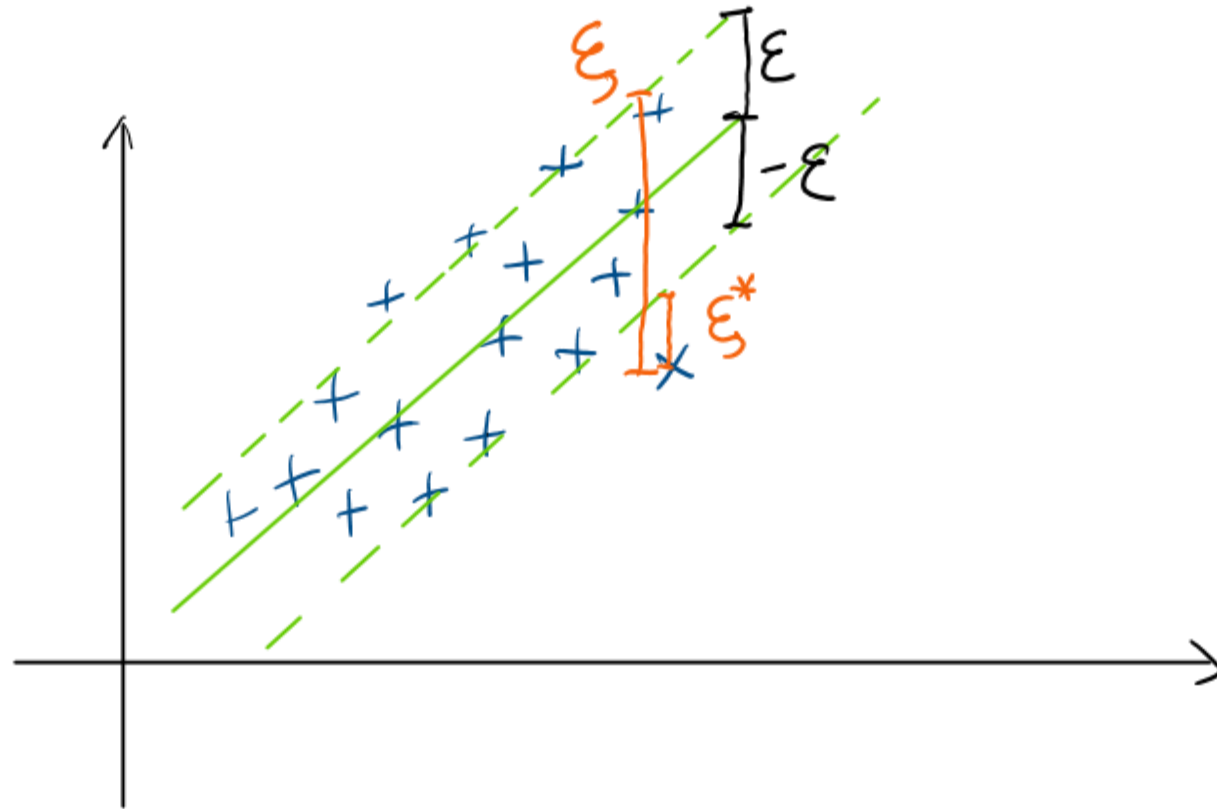
- Kernel SVM maps the original feature into a higher dimensional feature space.
- Use linear SVM to classify in the higher dimensional space.
- Use kernel trick to reduce the computational load that allows to calculate sample pair-wise dot product in the original feature space.

Polynomial Kernel: $k(x_i, x_j) = (x_i \cdot x_j + b)^m$

Gaussian Kernel: $k(x_i, x_j) = \exp\left(\frac{-\|x_i - x_j\|^2}{\sigma^2}\right)$



SVM for Regression



$$\min_{w, b, \xi, \xi^*} \frac{1}{2} \|W\|^2 + C \sum_{n=1}^N (\xi_n + \xi_n^*) \quad \text{s.t.} \quad |y_n - (w^T x_n + b)| < \xi + \xi_n$$



Questions: SVM

- What are the hyper parameters for soft SVM?
- What are the hyper parameters for kernel SVM?
- How to use SVM to solve multi-class classification? One-to-one
One-to-rest
- Soft SVM vs Linear Regression? Pros and Cons



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